UCS 503
Software Engineering Practice Project Report

CampusCare

An Integrated Digital Healthcare Platform
for Thapar's Health Center

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Chapter 1

Introduction

1.1 Background

CampusCare is an integrated digital healthcare platform designed for campus health centers. The system brings the major activities of a routine student healthcare visit together within a single platform.

In a conventional campus health-center environment, student registration, doctor queues, consultation records and prescriptions may be handled through separate or partially manual processes. This can make it difficult to maintain consistent records and can increase the amount of administrative work required from healthcare-center staff.

CampusCare addresses this problem through a centralized, role-based system. The platform connects student identification and check-in with doctor availability, queue management, consultation, medical history, prescription generation and medicine dispensing.

The system is divided into six major functional modules:

1. Authentication and User Management
2. Student Profile and Check-in
3. Doctor Availability and Queue Management
4. Consultation and Medical History
5. Prescription and Dispensing
6. Administration and Audit

These modules represent the main workflow of the system and are reflected in the Data Flow Diagrams and UML models presented later in this report.

1.2 Problem Statement

Campus health centers need an organized method for managing student visits, doctor availability, waiting queues, medical records and prescriptions.

When these activities are handled manually or through disconnected systems, several problems can occur:

- student information may have to be entered or verified repeatedly;
- managing separate doctor queues can become difficult;
- previous medical information may take time to retrieve;
- consultation and prescription records may become disconnected;
- dispensing information may not be clearly linked to the original prescription;
- administrative activities may be difficult to track.

CampusCare is intended to provide a centralized solution that manages the routine healthcare workflow from student check-in through consultation, prescription and dispensing.

1.3 Objectives

The main objectives of CampusCare are:

1. To provide secure authentication and role-based access.
2. To maintain student profile and identification information.
3. To provide an organized student check-in process.
4. To manage doctor availability and doctor-specific queues.
5. To provide authorized access to student medical history.
6. To allow doctors to record consultation information.
7. To allow doctors to create structured prescriptions.
8. To maintain separate prescription and dispensing records.
9. To provide administrative controls and audit information.
10. To provide a foundation for RFID/NFC-based identification where suitable hardware and browser support are available.

1.4 Scope

CampusCare focuses on the routine healthcare-management activities performed within a campus health center.

The core scope includes:

- authentication and user management;
- student profiles;
- student identification and check-in;
- doctor availability;
- doctor-specific queue management;
- consultation management;
- medical-history management;
- prescription generation;
- medicine dispensing;
- administration;
- audit information.

RFID/NFC-based identification, advanced real-time functionality, medication reminders and advanced reporting can be considered extensions to the core system.

Inventory management is outside the current project scope.

1.5 Users of the System

CampusCare provides different interfaces and permissions for the following roles.

1.5.1 Student

Students use the system for authentication, identification, check-in and accessing information relevant to their visits.

1.5.2 Doctor

Doctors manage their availability, access authorized student medical history, conduct consultations and create prescriptions.

1.5.3 Receptionist

Receptionists assist students with identification and check-in and manage routine queue-related operations.

1.5.4 Pharmacist

Pharmacists review valid prescriptions and record medicine dispensing.

1.5.5 Administrator

Administrators manage user accounts, system configuration, doctor availability, identification credentials and audit information.

Chapter 2

Requirements Analysis

2.1 Functional Requirements

The functional requirements of CampusCare are listed below.

ID	Requirement	Description
FR-01	Authentication	The system shall authenticate users and provide access according to their assigned role.
FR-02	Student Profile	The system shall maintain student profile and identification information.
FR-03	Student Check-in	The receptionist shall be able to verify and check in a student.
FR-04	Doctor Availability	The system shall maintain doctor availability information.
FR-05	Queue Management	The system shall maintain separate queues for different doctors.
FR-06	Queue Status	The system shall maintain queue position and status information.
FR-07	Medical History	An authorized doctor shall be able to access relevant student medical history.
FR-08	Consultation	The doctor shall be able to record the reason for consultation, diagnosis and clinical notes.
FR-09	Prescription	The doctor shall be able to create a structured prescription linked to the consultation.
FR-10	Dispensing	The pharmacist shall be able to record medicine dispensing against a valid prescription.
FR-11	Administration	The administrator shall be able to manage users, configuration and operational information.

FR-12	RFID/NFC	The system may support RFID/NFC-based identification where suitable hardware and browser support are available.
FR-13	Audit	Important administrative and system actions shall be recorded for auditing.
FR-14	No-show Handling	The queue system shall support handling of students who do not respond when called.
FR-15	Queue Redirection	Authorized staff shall be able to redirect a student to an appropriate doctor queue when required.

2.2 Non-Functional Requirements

2.2.1 Security

CampusCare shall restrict system functionality and healthcare information according to user roles. Authentication and authorization are therefore important parts of the system.

2.2.2 Reliability

Student, consultation, prescription and dispensing information should remain consistent and available during normal system operation.

2.2.3 Usability

The system should provide simple and understandable interfaces for students and healthcare-center staff.

2.2.4 Maintainability

The application should be modular so that individual components can be modified and maintained without unnecessary changes to unrelated components.

2.2.5 Performance

Routine operations such as login, check-in, queue operations and record retrieval should provide reasonable response times under normal campus health-center usage.

2.2.6 Auditability

Important administrative and system activities should be traceable through audit information.

2.3 Technology Stack

The CampusCare technology stack consists of:

- **Frontend:** React with Vite
- **Styling:** Tailwind CSS
- **Backend:** FastAPI
- **API:** REST API
- **Database:** MySQL
- **Database Management:** SQL with version-controlled migrations
- **Identification:** Staff-assisted check-in with planned RFID/NFC support
- **Testing:** Unit and integration testing

Chapter 3

System Design and Architecture

3.1 System Architecture

CampusCare follows a client-server architecture consisting of a frontend, backend and relational database.

The frontend is responsible for providing role-specific interfaces for students, doctors, receptionists, pharmacists and administrators.

The FastAPI backend handles authentication, authorization, business logic, queue operations, consultation processing, prescriptions and administrative operations.

MySQL provides persistent storage for the system data.

The overall architecture can be represented as:

React Frontend → FastAPI REST API → Application Logic → MySQL Database

This separation allows the user interface, application logic and database layer to be developed and maintained independently.

3.2 Context Diagram

The Context Diagram represents CampusCare as a single system and shows its interaction with the external actors.

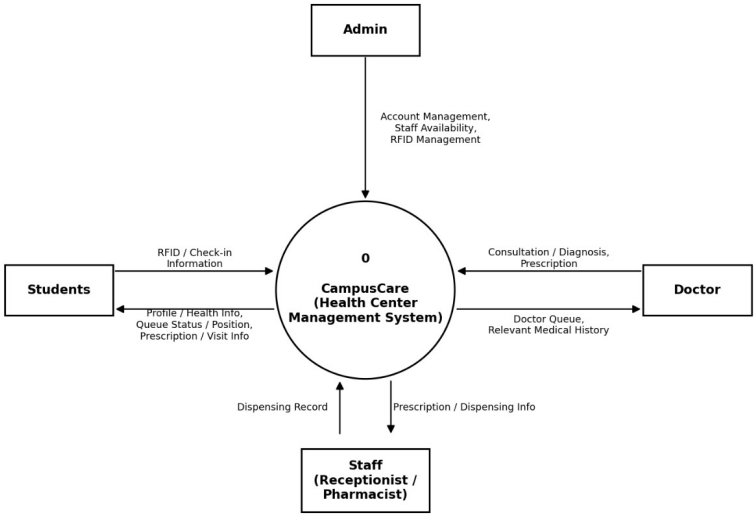


Figure 3.1: CampusCare Context Diagram

The main external actors are:

- Student
- Doctor
- Receptionist
- Pharmacist
- Administrator

The diagram provides a high-level view of the information exchanged between these actors and the CampusCare system.

3.3 Level-1 Data Flow Diagram

The Level-1 DFD decomposes CampusCare into six major processes.

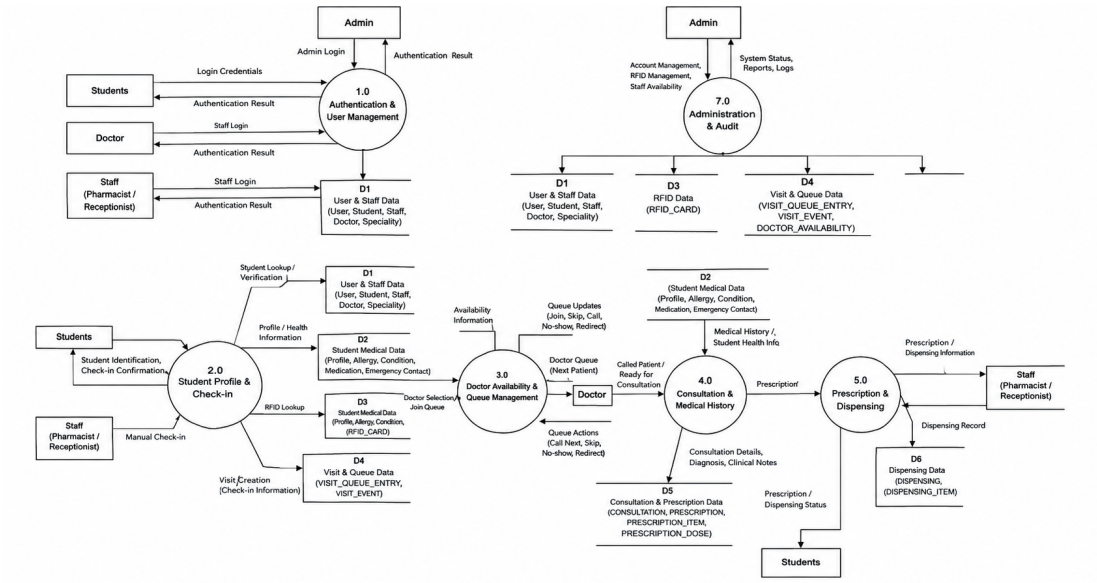


Figure 3.2: CampusCare Level-1 Data Flow Diagram

The six Level-1 processes are:

1. **1.0 Authentication and User Management**
2. **2.0 Student Profile and Check-in**
3. **3.0 Doctor Availability and Queue Management**
4. **4.0 Consultation and Medical History**
5. **5.0 Prescription and Dispensing**
6. **6.0 Administration and Audit**

The Level-2 diagrams below provide a more detailed decomposition of each of these processes.

3.4 Level-2 Data Flow Diagrams

3.4.1 1.0 Authentication and User Management

The first Level-2 DFD decomposes authentication and user management into smaller activities such as authentication, authorization and user-account management.

1.

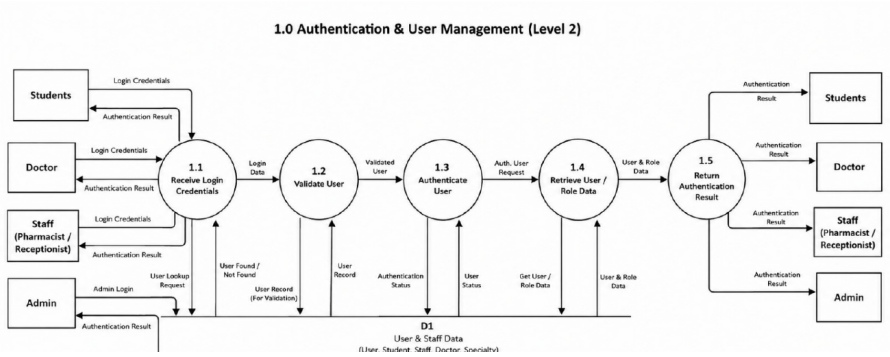


Figure 3.3: Level-2 DFD for Authentication and User Management

3.4.2 2.0 Student Profile and Check-in

This Level-2 DFD describes student identification, profile verification and the check-in process.

2.

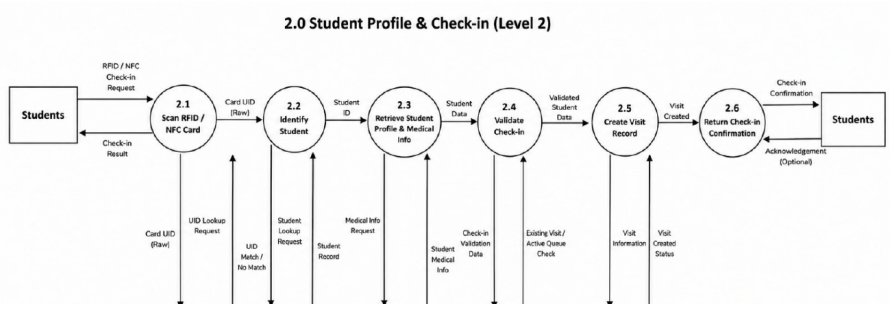


Figure 3.4: Level-2 DFD for Student Profile and Check-in

3.4.3 3.0 Doctor Availability and Queue Management

This Level-2 DFD describes doctor availability, queue creation, queue position, queue status, no-show handling and queue redirection.

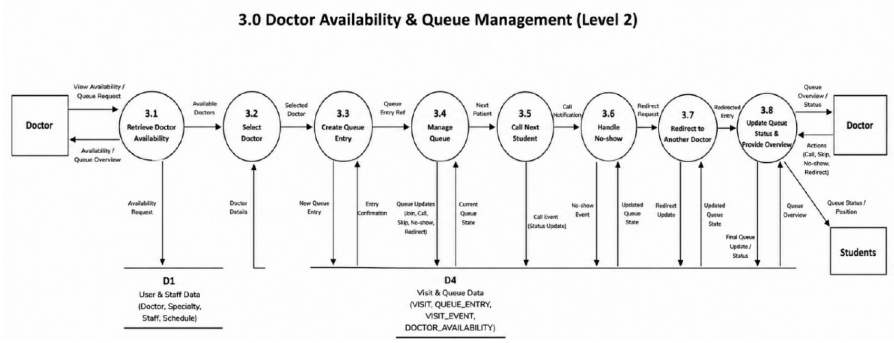


Figure 3.5: Level-2 DFD for Doctor Availability and Queue Management

3.4.4 4.0 Consultation and Medical History

This Level-2 DFD describes authorized access to student medical history and the recording of consultation information.

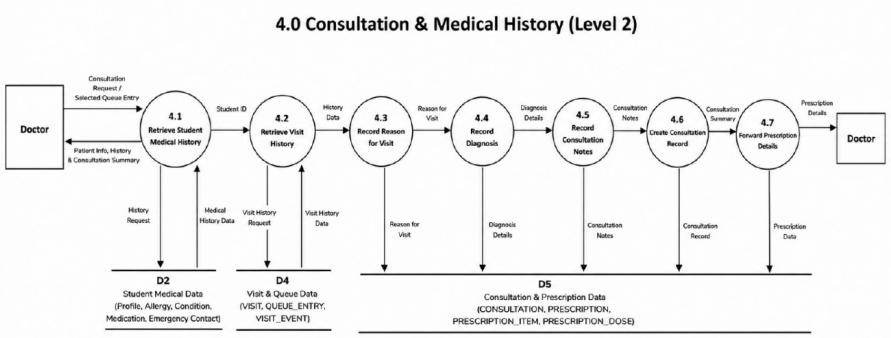


Figure 3.6: Level-2 DFD for Consultation and Medical History

3.4.5 5.0 Prescription and Dispensing

This Level-2 DFD describes prescription creation by the doctor and medicine dispensing by the pharmacist.

The prescription and dispensing records are kept separate. The prescription represents the doctor's original instruction, while the dispensing record represents the pharmacist's action.

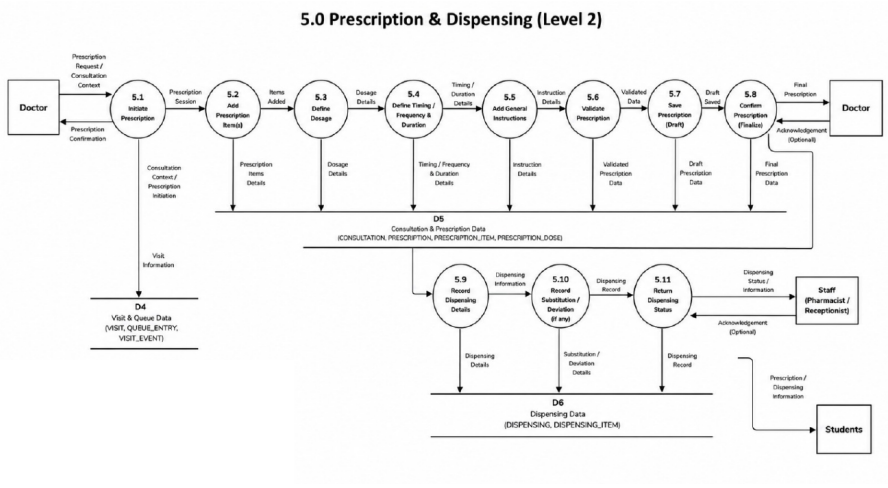


Figure 3.7: Level-2 DFD for Prescription and Dispensing

3.4.6 6.0 Administration and Audit

This Level-2 DFD describes administrative operations such as user management, doctor availability management, identification credential management, record retrieval and audit operations.

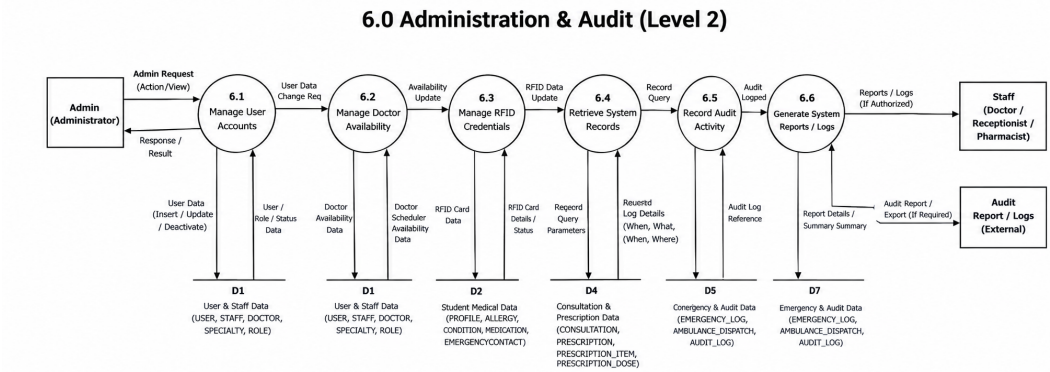


Figure 3.8: Level-2 DFD for Administration and Audit

3.5 UML Design

UML diagrams provide complementary views of the CampusCare system. Four UML diagrams are included:

1. Use Case Diagram
2. Activity Diagram
3. Class Diagram
4. Sequence Diagram

These diagrams describe the system from the functional, behavioral, structural and interaction perspectives.

3.5.1 Use Case Diagram

The Use Case Diagram represents the interactions between the different CampusCare actors and the system.

The main actors are Student, Doctor, Receptionist, Pharmacist and Administrator.

Major use cases include authentication, student check-in, queue management, consultation, medical-history access, prescription creation, medicine dispensing and administrative management.

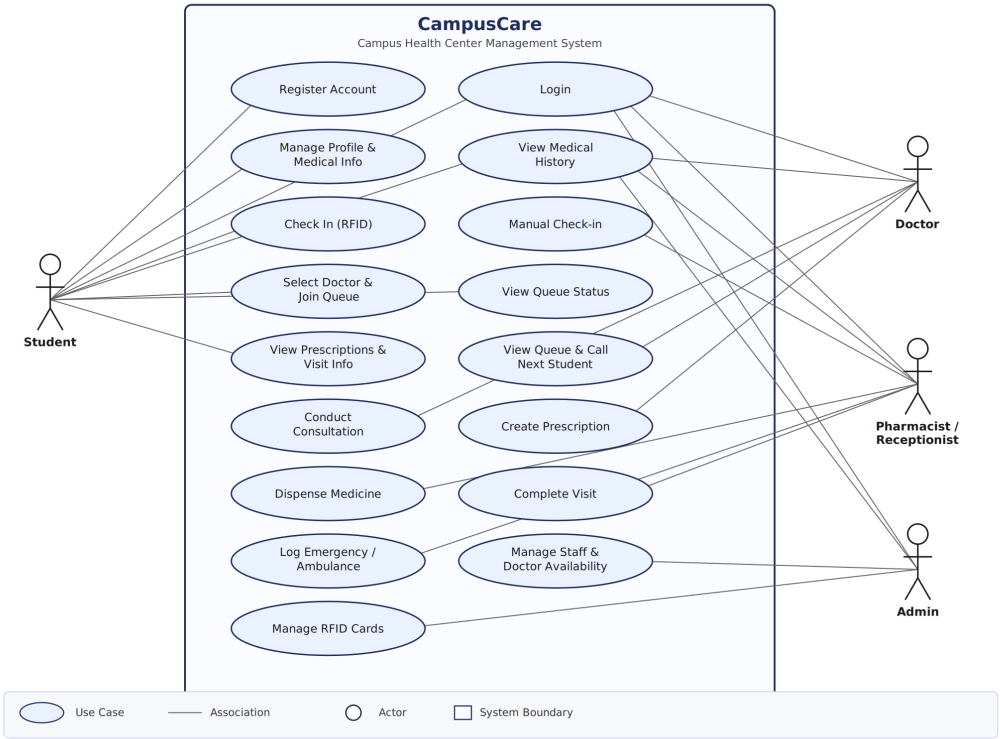


Figure 3.9: CampusCare Use Case Diagram

3.5.2 Activity Diagram

The Activity Diagram represents the normal workflow of a student health-center visit.

The process begins with authentication and student identification, followed by check-in, queue assignment, consultation and, when required, prescription and dispensing.

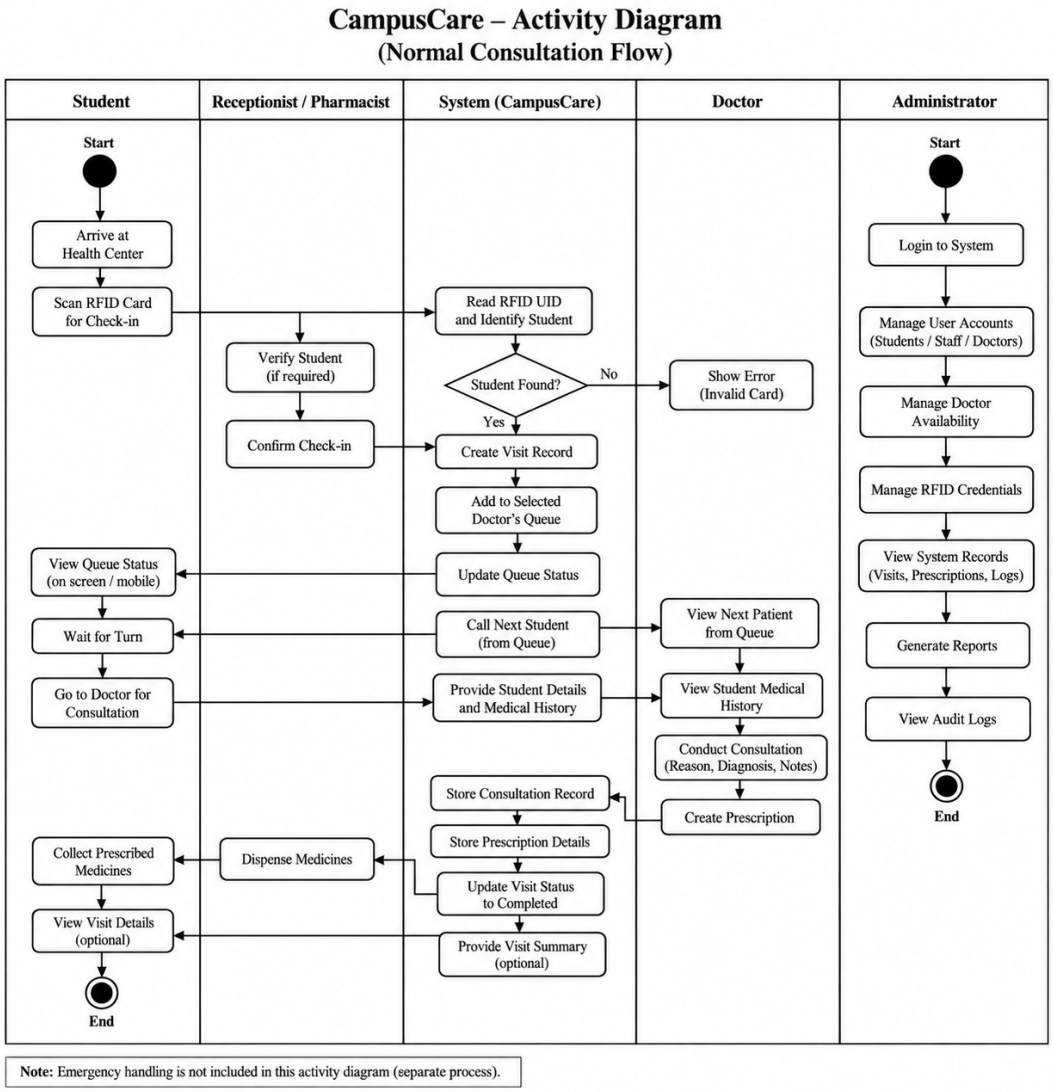


Figure 3.10: CampusCare Activity Diagram

The activity model focuses on the normal healthcare workflow and does not include any emergency or ambulance workflow.

3.5.3 Class Diagram

The Class Diagram represents the main structural entities of CampusCare and their relationships.

The model covers users and roles, student information, doctor availability, visits, queue entries, consultations, prescriptions, prescription items, dispensing records, RFID/NFC credentials and audit information.

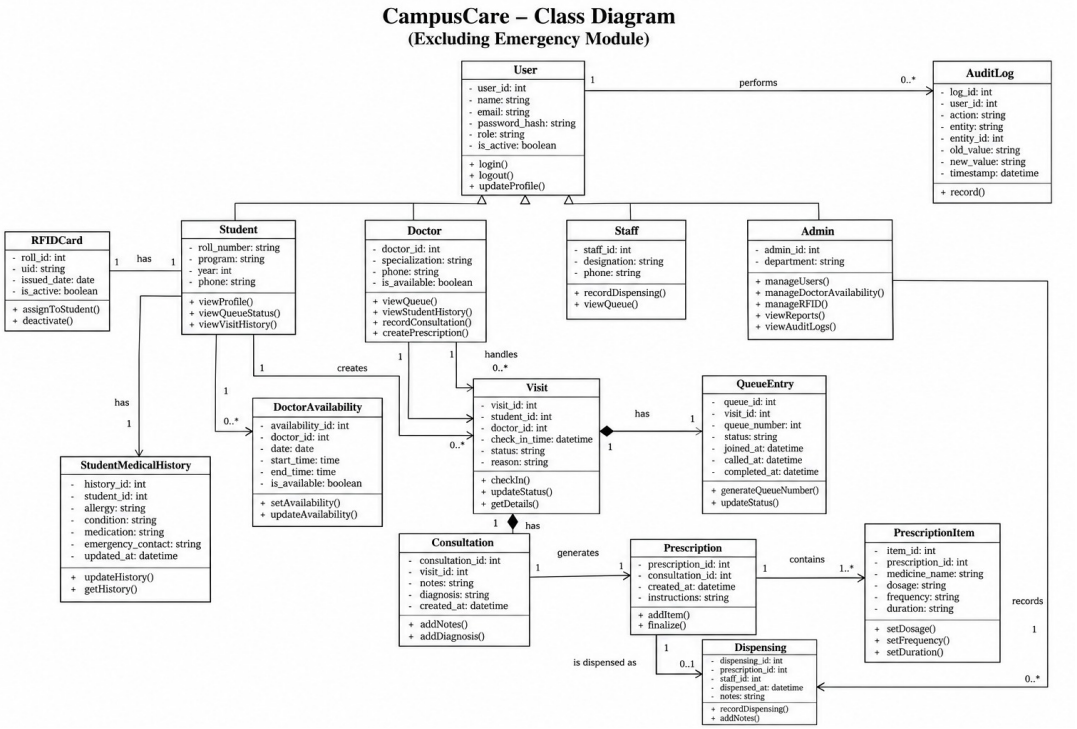


Figure 3.11: CampusCare Class Diagram

The class model is focused on the current CampusCare database and does not contain emergency or ambulance classes.

3.5.4 Sequence Diagram

The Sequence Diagram represents the interaction between the user, CampusCare application services and the MySQL database during a typical healthcare visit.

The sequence covers authentication, student check-in, queue processing, consultation, prescription creation and dispensing.

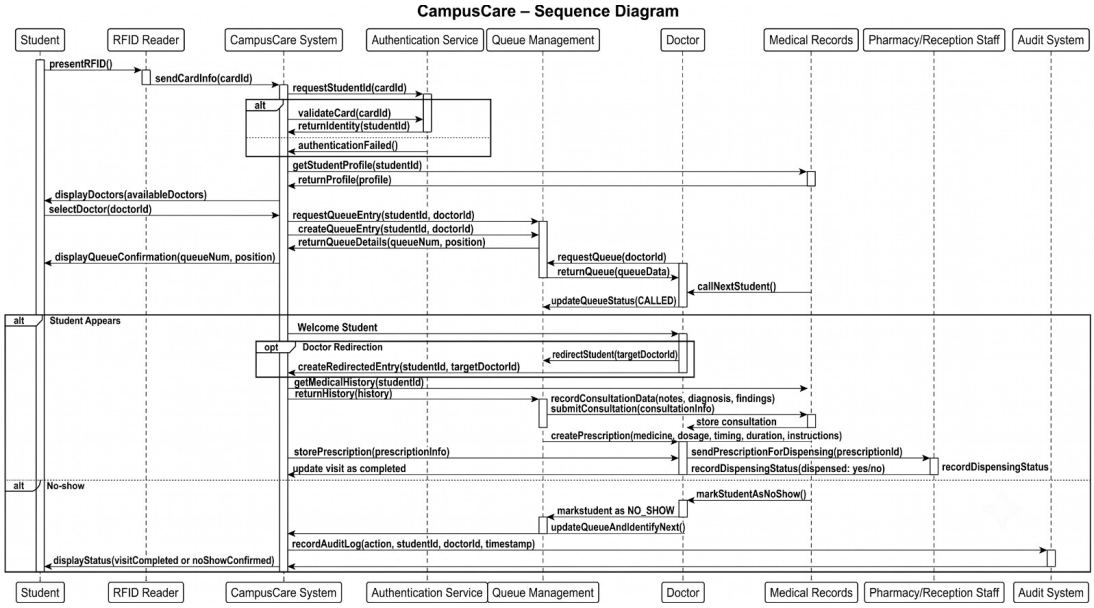


Figure 3.12: CampusCare Sequence Diagram

3.6 Database Design

CampusCare uses MySQL as its relational database.

The database is organized around the lifecycle of a routine student visit:

Student → Visit → Queue Entry → Consultation → Prescription → Dispensing

The major conceptual entities include:

- User
- Student
- Doctor
- Receptionist
- Pharmacist
- Administrator
- RFID Card
- Doctor Availability

- Visit
- Queue Entry
- Consultation
- Prescription
- Prescription Item
- Dispensing Record
- Audit Log

The exact implementation of these entities is maintained through the MySQL database schema and version-controlled migration files.

3.7 Prescription and Dispensing Separation

A major database-design decision is to keep prescriptions and dispensing as separate records.

A prescription records what the doctor has prescribed. A dispensing record records what the pharmacist has actually dispensed.

This separation prevents the dispensing process from silently changing the doctor's original prescription and provides a clearer record of the healthcare workflow.

3.8 Database Migration Strategy

Database schema changes are managed using version-controlled migration files.

This allows changes to the MySQL schema to be tracked and reproduced across development environments.

Chapter 4

System Implementation

4.1 Frontend Implementation

The CampusCare frontend is developed using React with Vite. Tailwind CSS is used for styling and interface development.

The frontend provides role-specific screens for the different CampusCare users.

4.1.1 Login Page

The login page provides the entry point for user authentication.

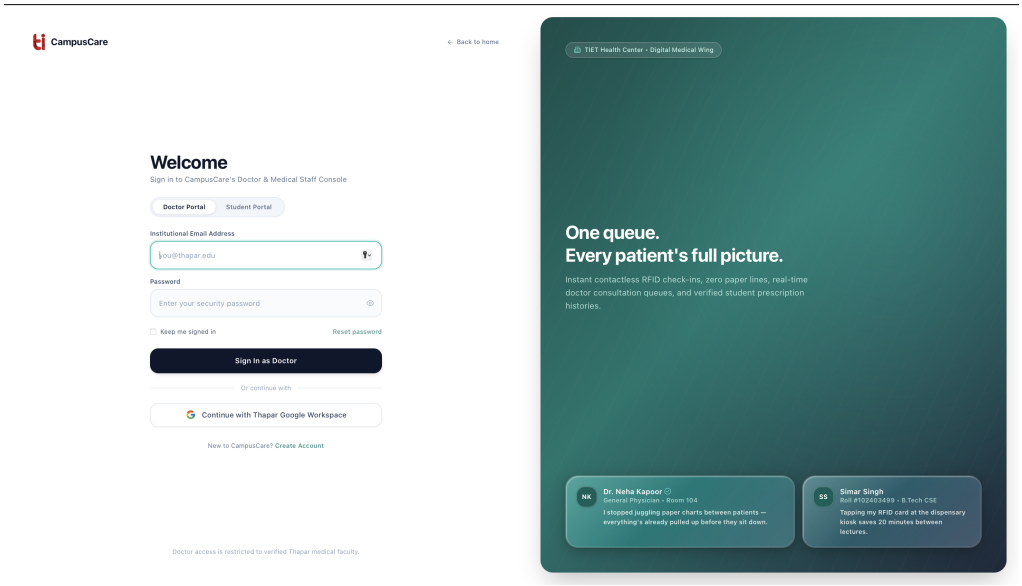


Figure 4.1: CampusCare Login Page

4.1.2 Landing Page

The landing page provides access to functionality available to the authenticated user.

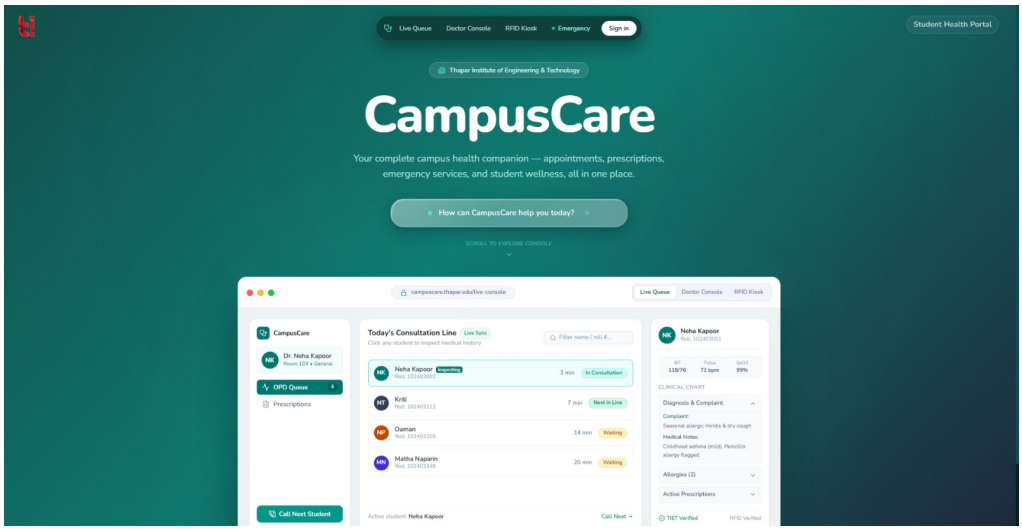


Figure 4.2: CampusCare Landing Page

4.1.3 Queue Management Console

The queue management console allows staff to monitor and manage doctor-specific queues.

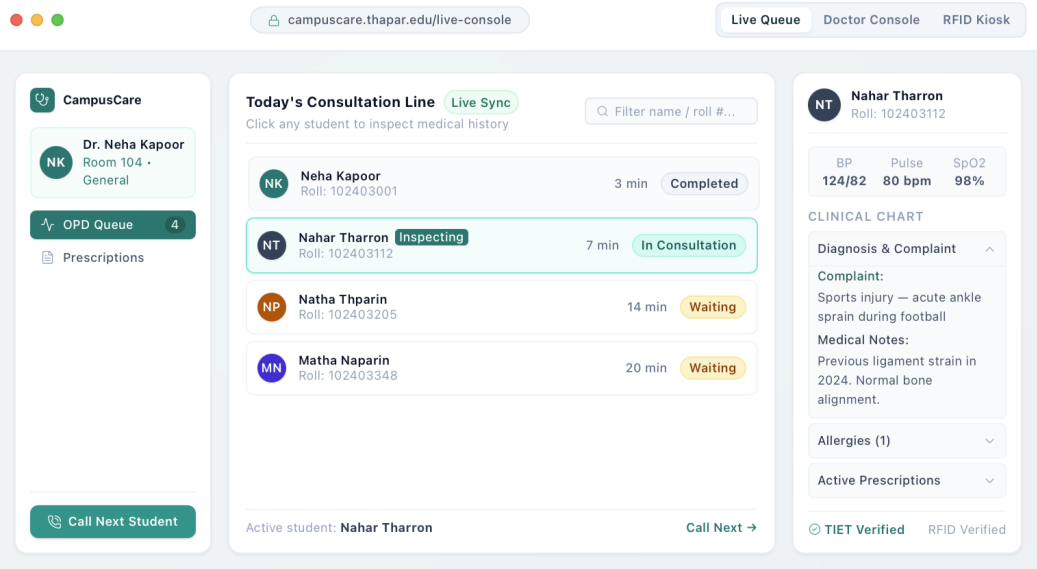


Figure 4.3: CampusCare Queue Management Console

4.1.4 Doctor Dashboard – Patient Record

The doctor dashboard provides access to authorized student records and relevant medical information.

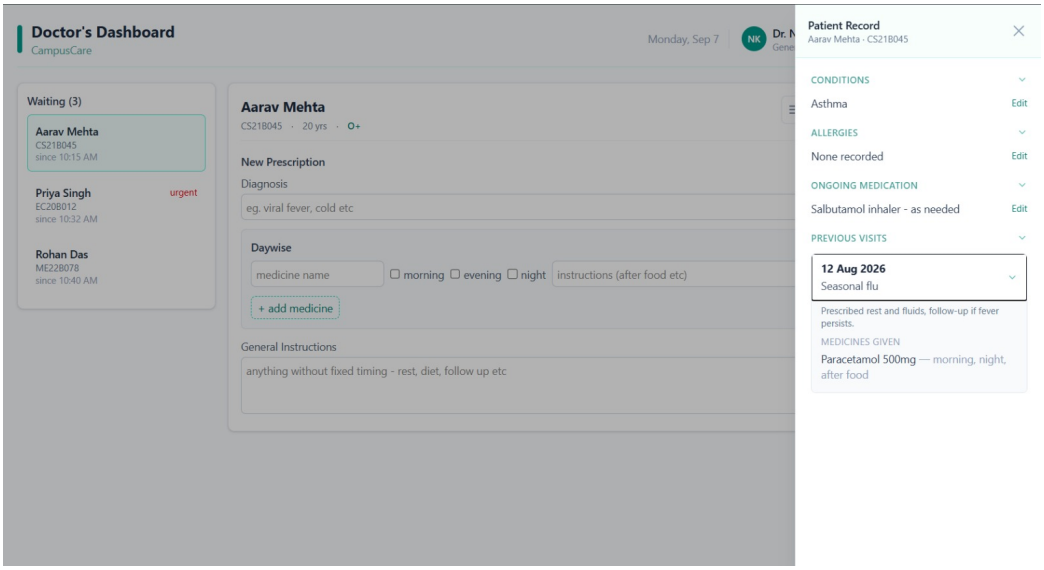


Figure 4.4: Doctor Dashboard – Patient Record

4.1.5 Doctor Dashboard – Prescription

The doctor dashboard also provides functionality for creating structured prescriptions.

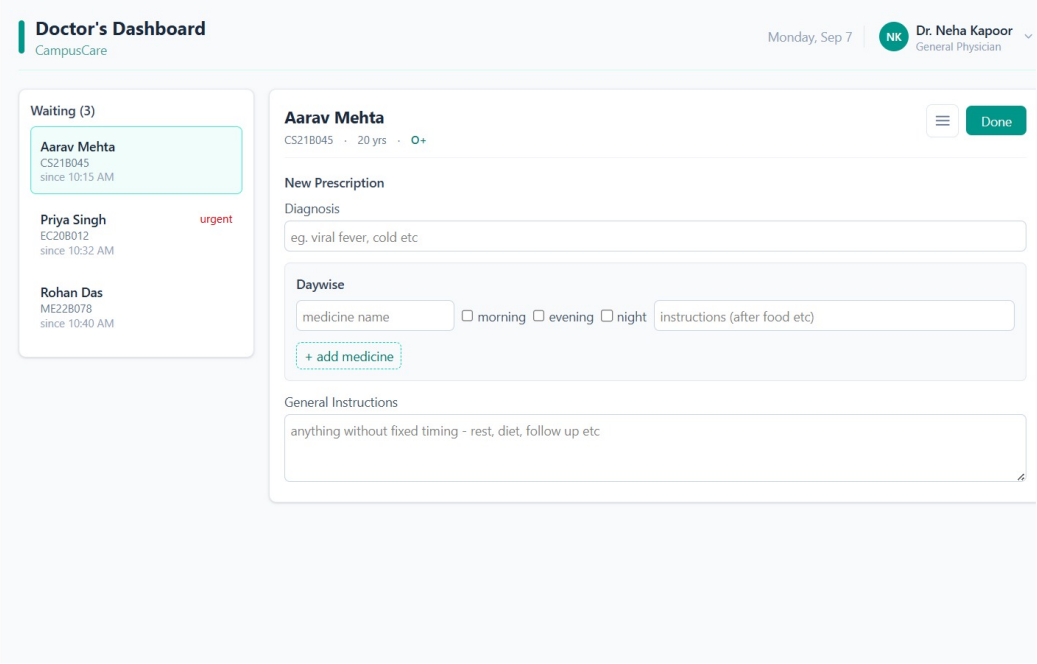


Figure 4.5: Doctor Dashboard – Prescription

4.2 Backend Implementation

The backend is implemented using FastAPI and exposes REST APIs for communication between the frontend and application services.

The backend is responsible for:

- authentication and authorization;
- user and role management;
- student profile management;
- student check-in;
- visit creation;
- doctor availability;
- queue management;
- medical-history access;
- consultation recording;

- prescription management;
- dispensing records;
- administrative operations;
- audit information.

4.3 Role-Based Access Control

Role-based access control separates the capabilities of students, doctors, receptionists, pharmacists and administrators.

This ensures that users are given access only to the functions required for their responsibilities.

Chapter 5

Project Development and Milestones

5.1 Development Phases

The development of CampusCare can be organized into the following phases.

Phase	Area	Major Work
1	Authentication and Profiles	Authentication, user roles and student profile management.
2	Check-in and Queue	Student check-in, doctor availability and queue management.
3	Consultation and Medical History	Medical-history access and consultation recording.
4	Prescription and Dispensing	Prescription creation and separate dispensing records.
5	Administration and Audit	Administrative controls, configuration and audit information.
6	Extensions	RFID/NFC support, advanced real-time functionality, reminders, analytics and additional reporting.

5.2 Core System Workflow

The normal CampusCare workflow is:

1. The user authenticates into the system.
2. The student's identity is verified.
3. The receptionist checks in the student.
4. A visit is created.
5. The student is added to the appropriate doctor's queue.
6. The doctor calls the next student.

7. The doctor accesses the student's authorized medical history.
8. The doctor conducts and records the consultation.
9. The doctor creates a prescription when required.
10. The pharmacist reviews the prescription.
11. The pharmacist records medicine dispensing.
12. The visit information remains available for future authorized access.

Chapter 6

Testing and Validation

6.1 Testing Strategy

CampusCare is tested through functional, integration and role-based testing. The main areas considered during testing are:

- authentication;
- student check-in;
- doctor availability;
- queue management;
- medical-history access;
- consultation recording;
- prescription creation;
- medicine dispensing;
- role authorization;
- administrative auditing.

6.2 Functional Test Cases

ID	Test Case	Expected Result	Status
TC-01	Valid Login	Valid credentials authenticate the user and provide the correct role-based interface.	To be recorded
TC-02	Invalid Login	Invalid credentials are rejected.	To be recorded
TC-03	Student Check-in	A valid student can be checked in by the receptionist.	To be recorded
TC-04	Queue Assignment	The student is added to the correct doctor's queue.	To be recorded

TC-05	Queue Status	The current queue position and status are displayed correctly.	To be recorded
TC-06	Doctor Availability	Doctor availability changes are reflected in queue operations.	To be recorded
TC-07	Medical History	An authorized doctor can access the relevant student medical history.	To be recorded
TC-08	Consultation	The doctor can record consultation information.	To be recorded
TC-09	Prescription	The doctor can create a structured prescription linked to the consultation.	To be recorded
TC-10	Dispensing	The pharmacist can record dispensing against a valid prescription.	To be recorded
TC-11	Role Authorization	Users cannot access functionality outside their assigned role.	To be recorded
TC-12	Audit	Important administrative actions are recorded correctly.	To be recorded

6.3 Validation

The test cases provide a structured way to validate the major functional requirements of CampusCare.

The status column is intentionally kept as “To be recorded” unless actual test results have been measured. This avoids presenting unverified testing results as completed results.

Chapter 7

Risks, Challenges and Mitigations

7.1 Security and Privacy

CampusCare handles student healthcare information, making controlled access important. Role-based authorization is used to limit access according to user responsibilities.

7.2 Queue Consistency

Queue information can become inconsistent if multiple users attempt to update the queue simultaneously.

The backend therefore acts as the authoritative source for queue state and operations.

7.3 RFID/NFC Dependency

RFID/NFC functionality depends on suitable hardware and browser or device support.

Staff-assisted identification remains available when RFID/NFC hardware is not available.

7.4 Data Integrity

Prescription and dispensing information are maintained separately.

This preserves the doctor's original prescription while allowing the pharmacist to record the dispensing action.

7.5 System Availability

CampusCare may be required during health-center operating hours. Therefore, back-end and database availability should be considered during deployment.

Chapter 8

Project Scope and Deliverables

8.1 Core Deliverables

The core CampusCare system includes:

- Authentication and role-based access.
- Student profile management.
- Staff-assisted student identification and check-in.
- Doctor availability management.
- Doctor-specific queue management.
- Queue position and status management.
- Medical-history access for authorized doctors.
- Consultation recording.
- Structured prescription creation.
- Separate dispensing records.
- Administrative management.
- Audit information.

8.2 Extended Functionality

The following features can extend the core system:

- RFID/NFC-based student identification;
- advanced real-time queue updates;
- additional queue-management functionality;
- additional operational reports.

8.3 Stretch Functionality

Possible future enhancements include:

- medication reminders;
- aggregated and de-identified health trends;
- advanced reporting and analytics;
- integration with additional campus healthcare services.

8.4 Out of Scope

The following functionality is outside the current project scope:

- inventory management;
- advanced healthcare analytics;
- external healthcare-system integration.

Chapter 9

Discussion and Limitations

9.1 Discussion

CampusCare connects the major routine healthcare-center activities within a single digital platform.

The system links student check-in and doctor queues with consultation records, prescriptions and dispensing. This creates a continuous workflow instead of treating each activity as an isolated process.

A key design decision is the separation between prescriptions and dispensing records. The prescription represents the doctor's instruction, while the dispensing record represents the pharmacist's action.

The DFD models describe how information moves through the system, while the UML diagrams provide different views of functionality, workflow, structure and interaction.

9.2 Limitations

The current project has several limitations:

- RFID/NFC functionality depends on available hardware and browser capabilities.
- Advanced real-time synchronization may require additional infrastructure.
- Quantitative performance measurements require controlled testing under realistic workloads.
- Advanced analytics are outside the current core scope.
- Inventory management is not included.

Chapter 10

Conclusion and Future Work

10.1 Conclusion

CampusCare provides a centralized approach to managing routine healthcare activities within a campus health center.

The system brings together authentication, student profiles, check-in, doctor availability, queue management, consultation, medical history, prescriptions, dispensing, administration and auditing.

Role-based access separates the responsibilities of students, doctors, receptionists, pharmacists and administrators.

The separation of prescriptions and dispensing records helps preserve the integrity of doctor-created medical instructions while maintaining a clear dispensing history.

The system is represented using Context and Data Flow Diagrams as well as UML Use Case, Activity, Class and Sequence Diagrams. Together, these models connect the requirements of CampusCare with its system design and implementation.

10.2 Future Work

Future development can include:

- broader RFID/NFC integration;
- improved real-time queue updates;
- medication reminders;
- aggregated and de-identified health analytics;
- advanced reporting;
- improved deployment and monitoring;
- integration with additional campus healthcare services.

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